

Corporate Bond Relative Value

What Changed Across Three Duration-Neutral Portfolio Passes

Course	MTH 9897 - Systematic Trading
Instructor	Professor Misha Boroditsky
Sample period	August 2022 to March 2025
Issuers	AAPL, AMZN, BA, CAT, DIS, and T

2.75% Pass 1 gross annualized return	4.48% Pass 2 gross annualized return	5.99% Pass 3 gross annualized return
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The result after three passes

Return improved as I removed weaker signals and then brought liquidity into the ranking. The final pass is the strongest one in this sample, but most of the improvement comes from AAPL and the last part of the test period. I therefore treat it as a useful portfolio-design result, not as a ready-to-trade return forecast.

Executive Summary

I began with a straightforward issuer-curve strategy, then changed one portfolio choice at a time. This made the improvement easier to understand: first I tested whether weaker residuals were diluting the trade, and then I tested whether liquidity should affect which bonds enter the portfolio.

The three passes

- Pass 1, the baseline top-three rule, earns 2.75% annualized with 4.47% volatility.
- Pass 2 keeps only the two strongest signals. Return rises to 4.48%, while volatility rises to 7.10%.
- Pass 3 restores three bonds per side but adjusts the ranking for TRACE dollar volume. It earns 5.99% gross and 3.60% after a 25 bp one-way turnover charge.
- Pass 3 has the best 25 bp cost-adjusted Sharpe ratio at 0.42, compared with 0.10 and 0.29 for the first two passes.

16.21% Gross total return	5.99% Gross annualized return	8.66% Annualized volatility	0.69 Gross Sharpe ratio
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How I read the result

The attractive finding is that the return and the cost-adjusted Sharpe ratio both improve by the third pass. The uncomfortable finding is that excluding AAPL reduces annualized return to 0.05% and Sharpe to 0.06. The three rules were also chosen and compared on the same sample. Those two checks keep me from interpreting the headline return as a stable expectation.

1. Data and Sample Construction

The analysis uses WRDS enhanced end-of-month corporate bond returns for six large non-financial issuers. The common sample begins in August 2022 and ends in March 2025. I require at least one year to maturity and approximately \$200 million outstanding, and I remove defaulted, convertible, and implausible observations.

Within each issuer-month, I keep the modal security level and rating category. That choice sacrifices some observations, but it makes the fitted curve a comparison among bonds with more similar seniority and credit quality.

Issuer	Observations	Unique bonds	Median duration	Duration range	Median spread
AAPL	1,475	52	7.31	0.95 - 20.99	19.0 bps
AMZN	908	33	6.58	0.95 - 20.03	15.0 bps
BA	1,304	51	7.39	0.92 - 16.41	28.5 bps
CAT	957	51	3.45	0.95 - 18.65	13.0 bps
DIS	1,237	44	8.82	0.96 - 18.19	20.0 bps
T	1,678	56	10.63	0.96 - 17.84	28.0 bps

Return treatment

Positions are formed at month-end and evaluated using the next observation for the same bond when the gap is between 25 and 45 days. WRDS RET_EOM is the primary total-return measure. If it is missing, the fallback adds clean-price return and coupon accrual using the actual day gap divided by 360. The fallback was available but was not needed for positions that entered this backtest.

Limits of the supplied fields

The files do not provide a complete call schedule, bid-offer spread, 144A flag, subsidiary mapping, or borrow availability. I keep those items as limitations instead of inferring them from incomplete fields.

2. Issuer Spread Curves

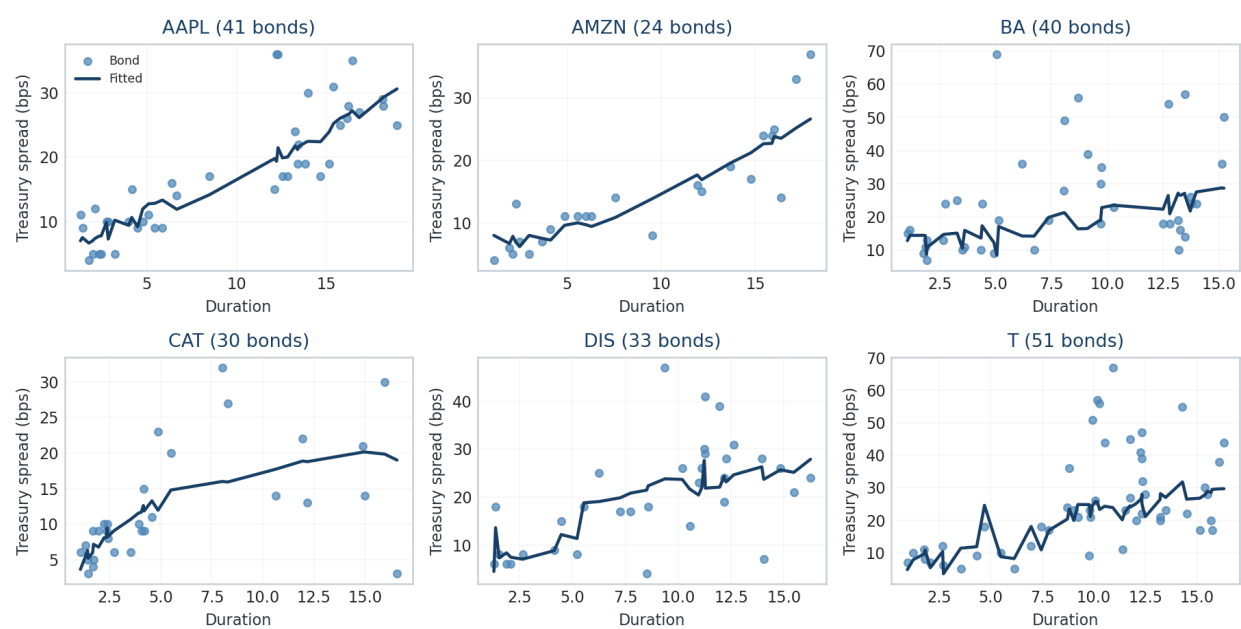
For each issuer-month, I estimate the following weighted regression:

Curve specification

$$T_Spread = b_0 + b_1 \log(1 + Duration) + b_2 Duration + b_3 Coupon$$

Professor Boroditsky emphasized that large, actively traded issues usually provide cleaner calibration points. I therefore combine TRACE dollar volume and amount outstanding in the regression weights. Both inputs are winsorized within the issuer-month so one unusually large bond cannot dominate the fit.

Issuer spread curves at March 2025



Issuer	Fit months	Average bonds	Median weighted R2	Median RMSE
AAPL	32	46.1	0.744	6.3 bps
AMZN	32	28.4	0.755	5.3 bps
BA	32	40.8	0.376	10.5 bps
CAT	32	29.9	0.617	7.1 bps
DIS	32	38.7	0.535	8.4 bps
T	32	52.4	0.386	12.8 bps

A positive residual means the observed spread is wider than the fitted curve and is classified as cheap. A negative residual is classified as rich.

3. Portfolio Construction

All three passes buy cheap bonds and short rich bonds. The short side is scaled so the signed market-value duration contribution is approximately zero, and gross market value is then normalized to one.

Hedge condition	$\text{sum}(\text{Position} \times \text{Price} \times \text{Duration}) = 0$
Equivalent DV01 condition	$\text{sum}(\text{Dollar DV01}) = 0$

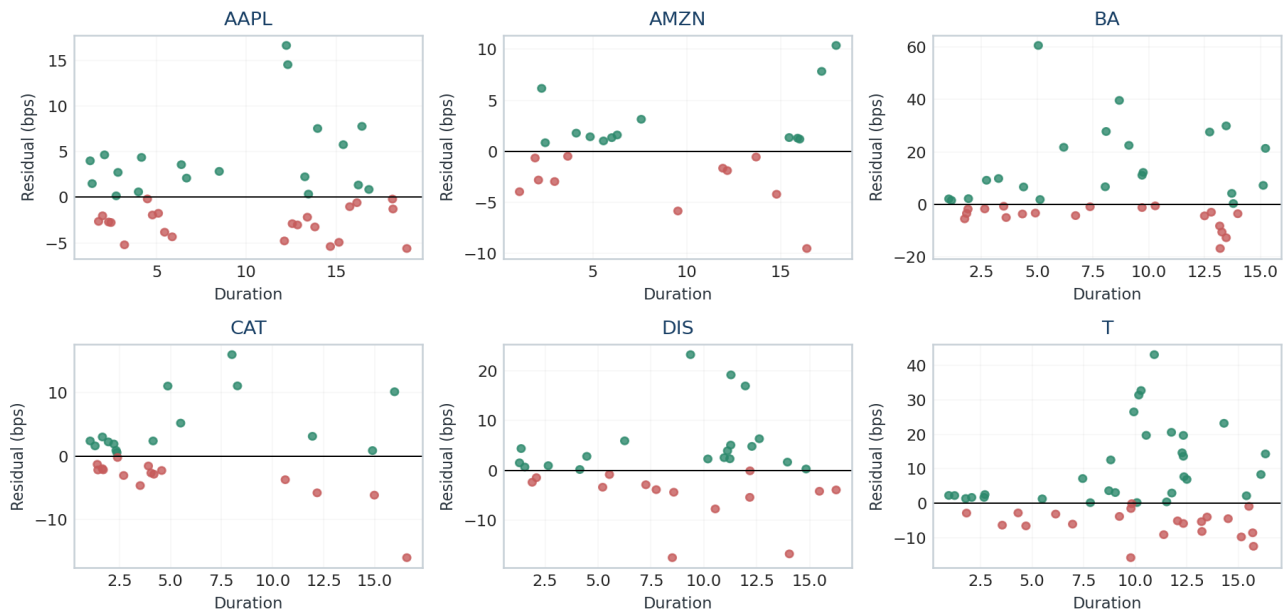
Why each pass changed

- Pass 1 establishes the baseline with three bonds on each side and position size proportional to residual strength.
- Pass 2 asks whether the weakest selected residuals dilute the payoff. It keeps only the two strongest signals on each side.
- Pass 3 addresses implementation. It keeps three bonds but gives more ranking weight to issues with stronger TRACE dollar volume.

Risk that remains

DV01 neutrality only addresses a first-order parallel rate move. The portfolio can still carry Treasury-curve slope and curvature risk, spread-duration risk, issuer beta, downgrade risk, liquidity exposure, and shorting costs.

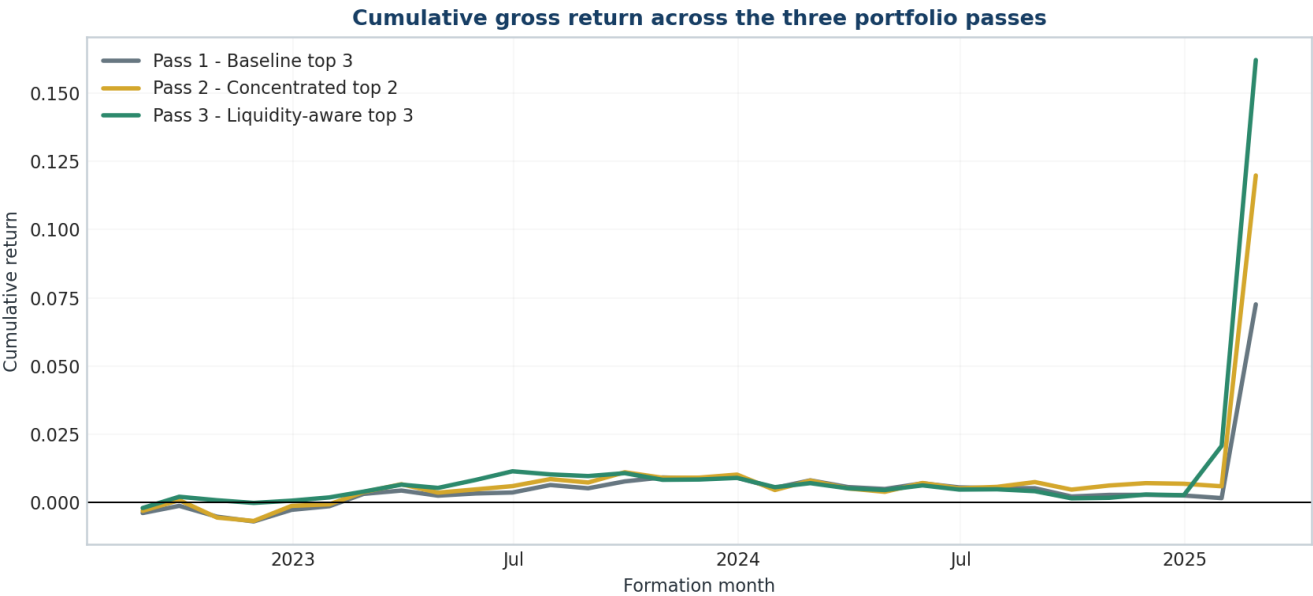
Rich and cheap residuals at March 2025



4. What Improved Across the Three Passes

The table uses the same 31-month evaluation window for all three versions. I select the final pass using the highest Sharpe ratio after a 25 bp one-way turnover charge rather than gross return alone.

Portfolio pass	Gross ann. return	Gross vol.	Gross Sharpe	Avg. turnover	Net ann. return (25 bp)	Net Sharpe (25 bp)
1. Baseline top 3	2.75%	4.47%	0.62	0.78	0.45%	0.10
2. Concentrated top 2	4.48%	7.10%	0.63	0.82	2.02%	0.29
3. Liquidity-aware top 3	5.99%	8.66%	0.69	0.79	3.60%	0.42



The progression is economically sensible: stronger signals improve the second pass, and liquidity-aware ranking improves the third. The cumulative chart also shows that much of the separation arrives near the end of the sample, which makes the result less convincing than the full-period return alone suggests.

What I would do differently in a live research process

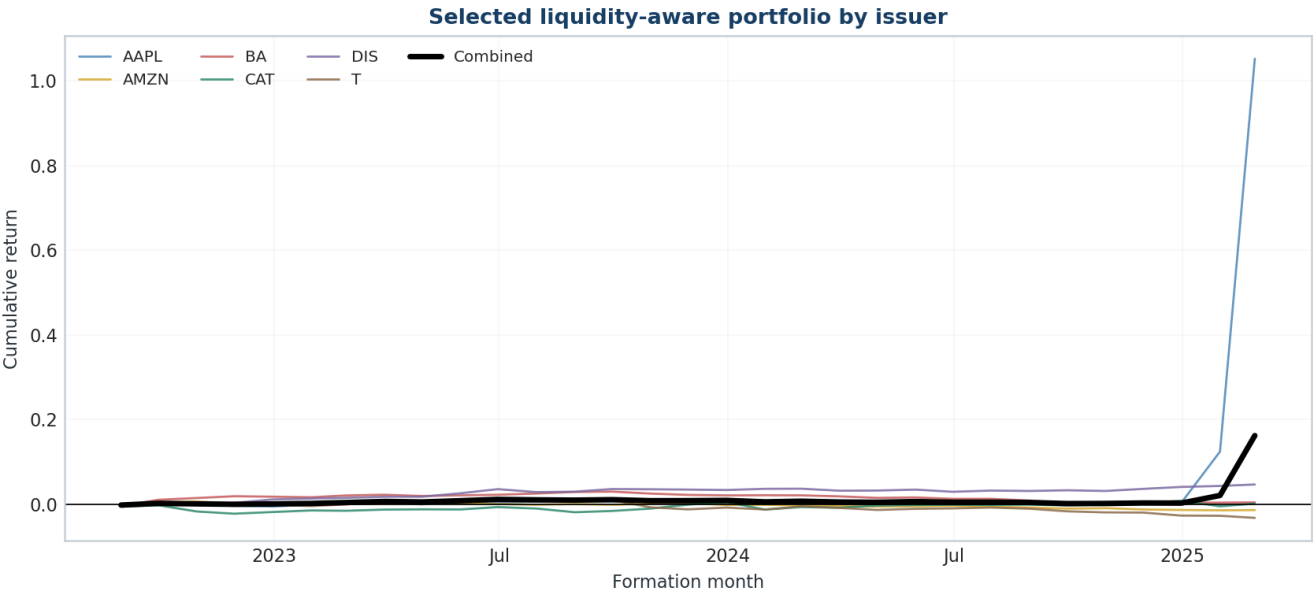
I would choose the portfolio rule using an earlier training period and leave the later observations untouched. Here, all three passes are compared on the same sample, so selecting the winner introduces model-selection bias.

5. Selected Portfolio Performance

The liquidity-aware top-three portfolio earns 5.99% gross annualized return with 8.66% volatility and a 0.69 Sharpe ratio. On the \$1 million gross-capital convention, the sum of monthly gross PnL is \$159,240.

5.99% Gross annualized return	8.66% Annualized volatility	0.69 Gross Sharpe ratio	\$159,240 Sum of monthly gross PnL
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Issuer	Ann. return	Ann. volatility	Sharpe	Total return	Win rate	Avg. turnover
AAPL	32.08%	51.62%	0.62	105.19%	54.8%	0.86
AMZN	-0.54%	0.67%	-0.81	-1.38%	45.2%	0.77
BA	0.17%	1.36%	0.13	0.44%	54.8%	0.75
CAT	0.06%	2.23%	0.03	0.15%	61.3%	0.77
DIS	1.78%	1.25%	1.42	4.67%	67.7%	0.84
T	-1.26%	1.78%	-0.71	-3.21%	35.5%	0.75

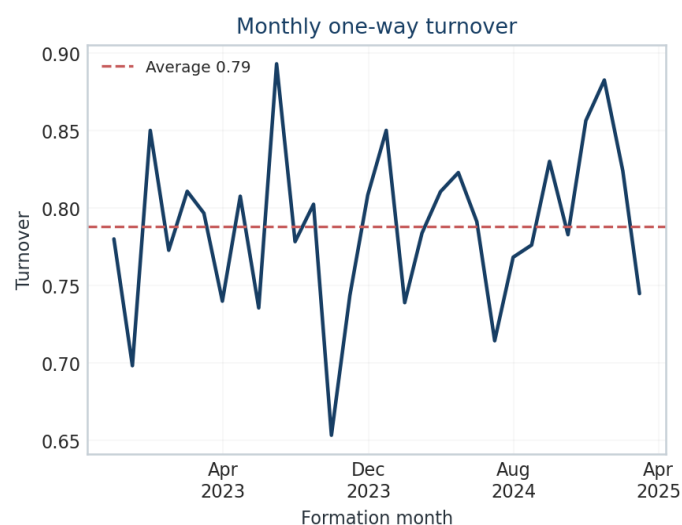
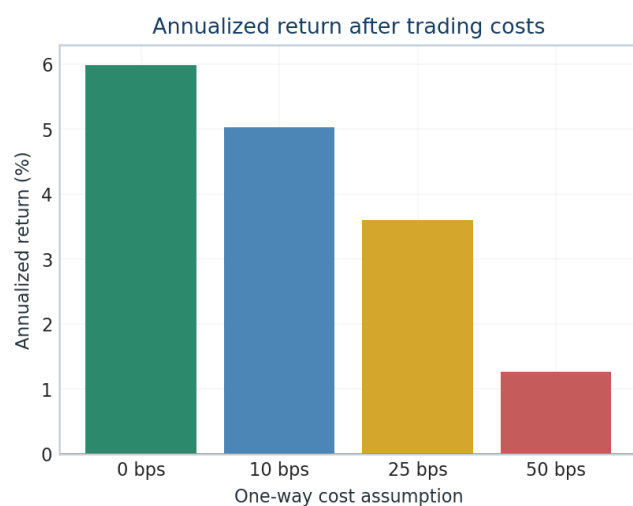


AAPL has an outsized effect, including a 13.84% return in the best month (2025-02-28). The worst combined month is 2024-01-31 at -0.34%.

6. Trading Costs and Robustness

Corporate-bond relative-value trades can have modest expected returns relative to bid-offer spreads, borrow costs, and market impact. I therefore apply several one-way cost assumptions to monthly one-way turnover.

Cost assumption	Ann. return	Ann. volatility	Sharpe	Total return	Win rate
0 bps	5.99%	8.66%	0.69	16.21%	54.8%
10 bps	5.03%	8.66%	0.58	13.51%	41.9%
25 bps	3.60%	8.66%	0.42	9.56%	22.6%
50 bps	1.26%	8.67%	0.15	3.28%	9.7%



Issuer concentration

Sample	Ann. return	Ann. volatility	Sharpe	Total return
All six issuers	5.99%	8.66%	0.69	16.21%
Excluding AAPL	0.05%	0.77%	0.06	0.12%

Removing AAPL reduces annualized return from 5.99% to 0.05% and Sharpe from 0.69 to 0.06. This was the check that changed my interpretation most: the final construction improves the headline return, but the gain is not broad across issuers.

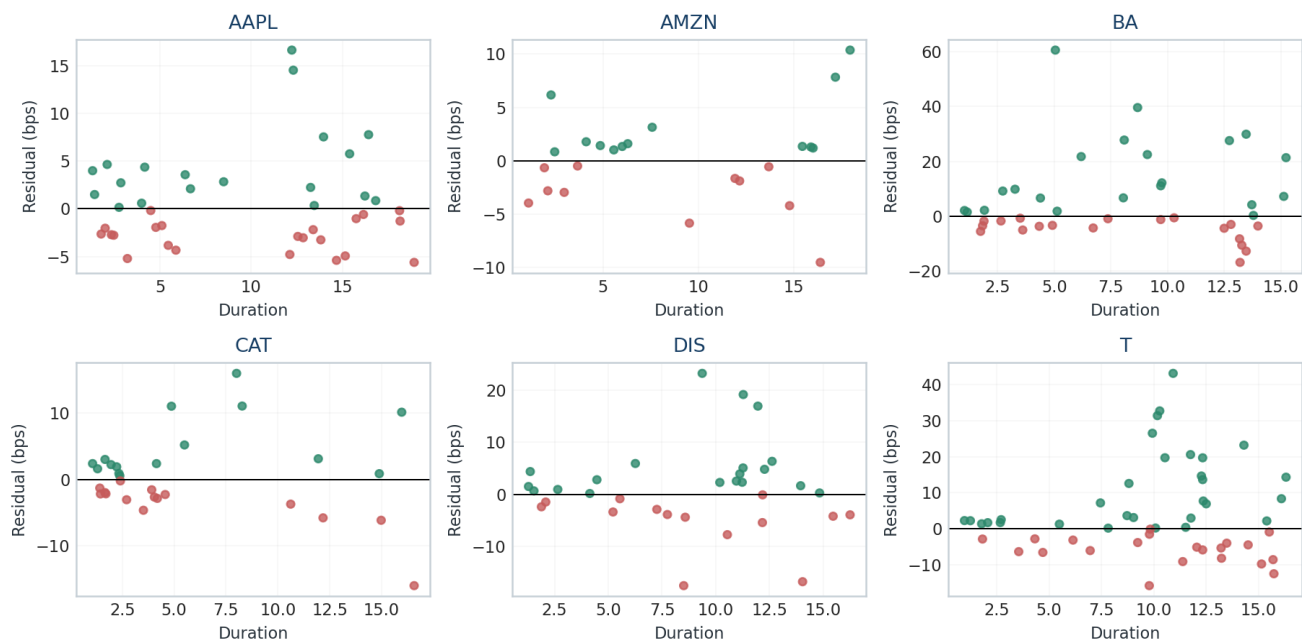
7. Outlier Review

In class, Professor Boroditsky discussed several reasons a bond can remain away from an issuer curve: liquidity, issue size, age, coupon structure, optionality, capital-structure position, and differences between the named issuer and the legal issuing entity. The supplied fields allow a partial check of those explanations.

Selection group	Observations	Median abs. residual	Median dollar volume	Median amount outstanding	Median duration	Median TMT
Long cheap	558	22.3 bps	\$35,557,574	\$750,000	12.32	18.95
Not selected	6,443	3.9 bps	\$45,349,053	\$1,000,000	6.86	8.24
Short rich	558	10.4 bps	\$109,797,947	\$1,750,000	13.21	23.75

The long-cheap observations have materially lower median dollar volume and smaller amount outstanding than the non-selected observations. They also have longer duration and time to maturity. My reading is that part of the cheap signal may compensate investors for liquidity and long-dated risk rather than represent a pure pricing error.

Rich and cheap residuals at March 2025



8. Conclusion

What I learned from the three passes is that the raw residual is only the starting point. Removing weaker signals raised the return, and adding liquidity to the ranking improved the cost-adjusted result again. The DV01 scaling also did what it was supposed to do mechanically: it removed first-order parallel-rate exposure.

I would still be cautious with the 5.99% headline return. Much of the improvement comes from AAPL and from the end of the sample. The long-cheap observations are also less liquid and longer-dated than bonds that were not selected. Those facts make the result interesting, but not yet stable enough to treat as a live expected return.

My conclusion is therefore fairly narrow: issuer-curve residuals are useful for organizing relative-value ideas, and portfolio construction can materially improve a backtest. A stronger next step would freeze the rule on an earlier sample and test it later with point-in-time bid-offer estimates, borrow constraints, and call information.

Assumptions and limitations

- The three portfolio passes are selected and evaluated on the same sample.
- The hedge neutralizes parallel-rate DV01, not all Treasury-curve or spread risk.
- Trading-cost cases do not include financing, borrow fees, failed fills, or asymmetric market impact.
- Call schedules, 144A status, subsidiary distinctions, insurance, and step-up coupons are not fully observed.
- Corporate bonds can be expensive or impossible to short, so the portfolio remains a research construction.

References

Boroditsky, M. MTH9897 Quantitative Investment Framework, class notes for sessions 1-2, 2026.

Nelson, C. R., and Siegel, A. F. (1987). Parsimonious Modeling of Yield Curves. *Journal of Business*, 60(4), 473-489.

WRDS Corporate Bond Returns, Enhanced End-of-Month files supplied for the course assignment.

Wharton Research Data Services. WRDS Bond Returns documentation:

<https://wrds-www.wharton.upenn.edu/pages/get-data/wrds-bond-returns/wrds-bond-returns/>